Fibocom
PERFECT WIRELESS EXPERIENCE

LogCapture Guide_Linux

V2.2

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Change History

V2.2 (2025-02-26)	Added the Samsung S5300 platform log grabbing guide. Added description of the maximum value of platform subcontracting for Unisoc UDX710/SL8563 in chapter 4. Added timeout settings for the TFTP tool in chapter 4. Added the description of tcpdump log capture mode on the cp side of the Samsung platform in chapter 4.
V2.1 (2024-06-04)	Added log capture guide for PCIE devices on the Unisoc platform.
V2.0 (2024-05-30)	Optimize some descriptions in the documentation
V1.9 (2024-05-11)	Added detailed usage descriptions of some platforms
V1.8 (2024-01-04)	Add the AP dump log capturing function on the UDX710/SL8563 Unisoc platform.
V1.7 (2023-12-20)	Optimize content
V1.6 (2023-12-05)	Added the description of grabbing logs in proxy mode
V1.5 (2023-11-14)	1. Add the dump log capturing function on the Qualcomm platform 2. Add the proxy log capture function
V1.4 (2023-10-28)	Add the description of the –z parameter
V1.3 (2023-09-23)	Add the description of the –a parameter
V1.2 (2023-05-11)	Add the description of the -a /–i/–u/–w/–m parameters
V1.1 (2022-12-16)	Add the description of the –h/–f parameters
V1.0 (2022-12-12)	Initial version

1 Forward

Logtool is used for Linux/Android hosts to obtain logs from the Fibocom module via USB.

1.1 Applicable Models

Table 1. Applicable model

Num	Modules	Description
1	L716/L718	ZTE V3E/T
2	L61x/LC61x/MC61x/LG61x	UNISOC 8910
3	MC66x/MG66x	UNISOC 8850
4	FG650,FG652	UNISOC UDX710
5	FG621	UNISOC SL8563
6	NL668/ MC116/LC116	QCOM MDM9x07
7	MA510/ MC109/MC100E	QCOM MDM9205
8	FM101/FG101	QCOM X12
9	FM150/FG150	QCOM X55
10	FM160/FG160	QCOM X62
11	LE170/LE270	EC618/EC718
12	FM190/FG190	QCOM X72
13	FG131/FG132	QCOM X35
14	FM170/FG170	QCOM X65
15	FM180/FG180	QCOM X72
16	FM550	SAMSUNG S5300

1.2 Adaptation platform

Linux platform/Android platform

2 Compilation

2.1 Tool source code introduction

- doc: Chinese and English guidance documents
- misc_code: common code
- qcom_code: Qcom modules code
- zte_code: ZTE modules code
- unisoc_code: Unisoc modules code
- samsung_code： Samsung module code
- eigencomm_code: Eigencomm modules code
- sl8563_x_code: Unisco UDX710/SL8563 platform related code
- main.c: The program code
- Makefile: Compiling configuration file for Linux Environment

The detailed directory structure is as shown in the figure 1:

```
├── logtool
├── main.c
├── Makefile
├── misc_code
│   ├── fifo.c
│   ├── fifo.h
│   ├── log_control.c
│   ├── log_control.h
│   ├── misc.c
│   ├── misc.h
│   ├── misc_usb.c
│   └── misc_usb.h
├── qcom_code
│   ├── config
│   │   ├── config_to_h_tool
│   │   │   ├── config_to_h_tool
│   │   │   ├── main.c
│   │   │   └── Makefile
│   │   ├── qxdm_default.cfg
│   │   └── qxdm_default_cfg.h
│   ├── ftp_client.c
│   ├── ftp_client.h
│   ├── qcom_devices_list.h
│   ├── qcom_log_main.c
│   ├── qlog.c
│   ├── qlog.h
│   ├── qshrink4.c
│   └── qshrink4.h
├── README
├── unisoc_code
│   ├── unisoc_devices_list.h
│   └── unisoc_log_main.c
├── unisoc_x_code
│   ├── unisoc_x_devices_list.h
│   ├── unisoc_x_log_main.c
│   └── xlog
│       ├── inc
│       │   ├── BaseChan.h
│       │   ├── ChannelDataHandler.h
│       │   ├── confile.h
│       │   ├── crc.h
│       │   ├── DiagChan.h
│       │   ├── DiagChannel.h
│       │   ├── DiagPackage.h
│       │   ├── dlmalloc.h
│       │   ├── ExePathHelper.h
│       │   ├── IComnChannel.h
│       │   ├── ISpLog.h
```

- unisoc_code: 展锐模块
- main.c: 下载主程序代码
- Makefile: Linux 环境编
- Android.mk: Android

详细目录结构如图 1 所示:

Figure 1. Detail directory structure

2.2 Compile Tool

2.2.1 Linux compilation

- Configure cross compilation tool

GCC is used by default. If arm GCC is required, set CROSS_COMPILE variable in Makefile.

As blow:

```
# If use the arm gcc, please set the arm gcc path.
# ARM_GCC = ~/ARM_Linux_GCC/bin/arm-none-linux-gnueabi-

CC = $(ARM_GCC)gcc
CXX = $(ARM_GCC)g++
AR = $(ARM_GCC)ar
CFLAGS = -Wall

LD_LIB = -lpthread -lxlog

INCLUDE = -I./misc_code -I./zte_code -I./qcom_code -I./unisoc_code -I./zte_code/config -I./qcom_code/config
INCLUDE += -I./unisoc_x_code/xlog/inc -I./unisoc_x_code/xlog/lib/inc
LIB_PATH = -L./unisoc_x_code/xlog/lib
SRC_FILES = $(wildcard ./misc_code/*.c ./zte_code/*.c ./qcom_code/*.c ./unisoc_code/*.c ./*.c)
SRC_FILES += $(wildcard ./unisoc_x_code/*.c)
SUBDIRS := ./unisoc_x_code/xlog

BIN_TARGET = logtool

OBJECTS = $(patsubst %.c,%.o,$(SRC_FILES))

$(BIN_TARGET) : $(OBJECTS)
>---make -C $(SUBDIRS) CC=$(CC) CXX=$(CXX) AR=$(AR)
>---$(CXX) $^ -o $@ $(LD_LIB) $(LIB_PATH)
>---@rm -f ./misc_code/*.o ./qcom_code/*.o ./zte_code/*.o ./unisoc_code/*.o ./*.o
>---@rm -f ./unisoc_x_code/*.o

$(OBJECTS) : %.o : %.c
>---$(CC) -c $< -o $@ $(CFLAGS) $(INCLUDE)

.PHONY: clean
```

Figure 2. Cross-compilation tool

- Compile

Put the Logtool code on the Linux host, then in the code directory, execute make. The Logtool will be generated if the compilation is OK.

As shown in the figure 3.

```
fengxianglong@Vostro-3267:~/Joy/Test/Fibocom_Linux_LogCapture$ ls
Android.mk  main.c  Makefile  misc_code  qcom_code  README  unisoc_code  unisoc_x_code  zte_code
fengxianglong@Vostro-3267:~/Joy/Test/Fibocom_Linux_LogCapture$ make
gcc -c misc_code/log_control.c -o misc_code/log_control.o -Wall -g -I./misc_code -I./zte_code -I./qcom_code -I./unisoc_co
gcc -c misc_code/misc.c -o misc_code/misc.o -Wall -g -I./misc_code -I./zte_code -I./qcom_code -I./unisoc_code -I./zte_coo
gcc -c misc_code/fifo.c -o misc_code/fifo.o -Wall -g -I./misc_code -I./zte_code -I./qcom_code -I./unisoc_code -I./zte_coo
gcc -c misc_code/misc_usb.c -o misc_code/misc_usb.o -Wall -g -I./misc_code -I./zte_code -I./qcom_code -I./unisoc_code -I.
gcc -c zte_code/zte_log_main.c -o zte_code/zte_log_main.o -Wall -g -I./misc_code -I./zte_code -I./qcom_code -I./unisoc_co
gcc -c qcom_code/qlog.c -o qcom_code/qlog.o -Wall -g -I./misc_code -I./zte_code -I./qcom_code -I./unisoc_code -I./zte_coo
gcc -c qcom_code/qcom_log_main.c -o qcom_code/qcom_log_main.o -Wall -g -I./misc_code -I./zte_code -I./qcom_code -I./unisoc
gcc -c qcom_code/qshrink4.c -o qcom_code/qshrink4.o -Wall -g -I./misc_code -I./zte_code -I./qcom_code -I./unisoc_code -I.
gcc -c qcom_code/ftp_client.c -o qcom_code/ftp_client.o -Wall -g -I./misc_code -I./zte_code -I./qcom_code -I./unisoc_code
gcc -c unisoc_code/unisoc_log_main.c -o unisoc_code/unisoc_log_main.o -Wall -g -I./misc_code -I./zte_code -I./qcom_code -
gcc -c main.c -o main.o -Wall -g -I./misc_code -I./zte_code -I./qcom_code -I./unisoc_code -I./zte_code/config -I./qcom_co
gcc -c unisoc_x_code/unisoc_x_log_main.c -o unisoc_x_code/unisoc_x_log_main.o -Wall -g -I./misc_code -I./zte_code -I./qco
g++ misc_code/log_control.o misc_code/misc.o misc_code/fifo.o misc_code/misc_usb.o zte_code/zte_log_main.o qcom_code/qlog
gtool -lpthread -lxlog  -L./unisoc_x_code/xlog/lib
fengxianglong@Vostro-3267:~/Joy/Test/Fibocom_Linux_LogCapture$ ls
Android.mk  logtool  main.c  Makefile  misc_code  qcom_code  README  unisoc_code  unisoc_x_code  zte_code
```

Figure 3. Linux compile

2.2.2 Android compilation

Decompress the tool codes and place them in the Android compilation environment, configure the environment variables, as shown in the following figure:

```
wujiabao@:~/Android/android11-rk3568$
wujiabao@:~/Android/android11-rk3568$ source build/envsetup.sh
wujiabao@:~/Android/android11-rk3568$
wujiabao@:~/Android/android11-rk3568$ lunch

Lunch menu... pick a combo:
     1. PX30_Android11-user
     2. PX30_Android11-userdebug
     3. aosp_arm-eng
     4. aosp_arm64-eng
     5. aosp_blueline_car-userdebug
     6. aosp_bonito_car-userdebug
     7. aosp_car_arm-userdebug
     8. aosp_car_arm64-userdebug
     9. aosp_car_x86-userdebug
     10. aosp_car_x86_64-userdebug
     11. aosp_coral_car-userdebug
     12. aosp_crosshatch_car-userdebug
     13. aosp_flame_car-userdebug
     14. aosp_x86-eng
     15. aosp_x86_64-eng
     16. car_x86_64-userdebug
     17. qemu_trusty_arm64-userdebug
     18. rk3288_Android10-user
     19. rk3288_Android10-userdebug
```

Figure 4. Configuring environment variables

Enter the tool code directory and execute mm for compilation:

```
wujiabao@:~/Android/android11-rk3568/external/Fibocom_Linux_LogCapture_V1.3.0.6$ mm
============================================
PLATFORM_VERSION_CODENAME=REL
PLATFORM_VERSION=11
TARGET_PRODUCT=rk3568_firefly_aioj
TARGET_BUILD_VARIANT=user
TARGET_BUILD_TYPE=release
TARGET_ARCH=arm64
TARGET_ARCH_VARIANT=armv8-a
TARGET_CPU_VARIANT=cortex-a55
TARGET_2ND_ARCH=arm
TARGET_2ND_ARCH_VARIANT=armv8-2a
TARGET_2ND_CPU_VARIANT=cortex-a55
HOST_ARCH=x86_64
HOST_2ND_ARCH=x86
HOST_OS=linux
HOST_OS_EXTRA=Linux-4.4.0-31-generic-x86_64-Ubuntu-14.04.6-LTS
HOST_CROSS_OS=windows
HOST_CROSS_ARCH=x86
HOST_CROSS_2ND_ARCH=x86_64
HOST_BUILD_TYPE=release
BUILD_ID=RQ3A.210705.001
OUT_DIR=out
============================================
[ 22% 64/285] including external/rk_tee_user/v2/Android.mk ...
'building rk_tee_user v2'
[ 33% 96/285] including hardware/gps/wu_gps/Android.mk ...
hardware/gps/wu_gps/Android.mk:32: warning: TARGET_ARCH=arm64
hardware/gps/wu_gps/Android.mk:33: warning: PLATFORM_VERSION_CODENAME=REL
hardware/gps/wu_gps/Android.mk:34: warning: PLATFORM_VERSION=11
hardware/gps/wu_gps/Android.mk:37: warning: MAJOR_VERSION=11
hardware/gps/wu_gps/Android.mk:38: warning: MINOR_VERSION=11
hardware/gps/wu_gps/Android.mk:44: warning: -Wno-unused-variable -Wno-unused-parameter -Wno-unused-function -Wno-return-stack-address -DGHT_FEATURE_ANDROID7_PLUSX
[ 44% 126/285] including hardware/rockchip/camera_engine_rkisp/Android.mk ...
TARGET_BOARD_PLATFORM=rk356x
[ 46% 133/285] including hardware/rockchip/libhwjpeg/Android.mk ...

[ 47% 136/285] including hardware/rockchip/librga/Android.mk ...
RGA is compiling with Android.mk Generated version.h
[ 48% 138/285] including hardware/rockchip/omx_il/Android.mk ...
date: Unknown option 'R' (see "date --help")
cp: .git/hooks/pre-commit: No such file or directory
tools/hooks .git/hooks
[ 99% 284/285] finishing build rules ...
packages/apps/Camera2/Android.mk: warning: "Camera2 (app:platform) should not link against libjni_jpegutil (native:ndk:libc++:static)"
[100% 290/290] Install: out/target/product/rk3568_firefly_aioj/system/bin/log_tool
```

Figure 5. compile

3 Obtain Logs

3.1 Obtain Logs

1. Check whether the USB connection of the module is normal.

```
lsusb
```

> Note, if the qcom module is in normal mode, please send at+disk=0,0,0 to unlock diag port first. Qualcomm X35 needs to first send AT+GTDIAGEN=1,1 to unlock the diag port.

2. Copy the cross compiled logtool tool to the host. For Samsung platforms, you will need to copy the compiled logtool, dlt-logcat, and silentlogging files to the host.
3. Enter the directory where logtool is located, and execute the command to obtain logs (the tool needs to have execution permission and access to the device node, otherwise an error may be reported due to permission problems).

```
./ logtool
```

3.2 Viewing logs

1. If the log is obtained successfully, the directory and file name of the log will be printed
2. Log acquisition failed, the reason for log acquisition failure can be seen from the print information.
3. If the - s parameter is added when obtaining logs, the logtool tool will save the log files to the specified path.

4 Parameter Description

Table 2. Parameter introduction

No.	Parameter	Necessary	Description
1	-d diagport	Yes	Specify diag port For example:./logtool – d /dev/ttyUSB1, where /dev/ttyUSB1 is the diagport enumerated by the host computer Samsung platform must use this parameter, other platforms as required.
2	-s log save dir	No	Set the log save path. For example:./logtool – s /sdcard/, where /sdcard/ is the path where the host computer saves log files. It should be noted that the logtool needs to have read and write permissions for the files in this path
3	-m single logfile size(MB)	No	Set the size of a single log file(The UDX710/SL8563 platform applies only to cp logs. The default value is 500MB and the maximum value is 500MB.) For example:./logtool -m 50 -n 10， Where -m 50 represents a maximum file size of 50M for a single log file
4	-n max_log_filenum	No	Set the maximum number of log files. If the limit is exceeded, the oldest log files will be deleted circularly. (1. UNISOC8910 and UNISOC8850 platform are used to set the threashold of the save log folder,unit 256MB 2. The UDX710/SL8563 platform applies only to cp logs) For example:./logtool -m 50 -n 10， Among them, - n 10 indicates that the maximum number of total files in the log directory is 10
5	-p logport	No	The UDX710 platform specifies the CP log capture port (log port) For example:./logtool – p /dev/ttyUSB2, where /dev/ttyUSB2 is the cp log port enumerated by the host computer
6	-h	No	View tool usage help For example:./logtool -h

7	-f config_file	No	The QCOM platform specifies a configuration file (qxdm_default.cfg), which is used by default without parameters, and is specified when a special configuration file is required to obtain logs For example:./logtool -f ./audio_mask.cfg, Among them ./ audio_mask.cfg is the specific path and name of the specified configuration file. Samsung platform specifies the ap log filter file or cp log filter level (can be 0,1,2,3, default is 0).
8	-a <0/1/2>	No	UDX710/SL8563 Platform configuration -a 0 ap+cp logs are captured locally. The -a 2 command is used to capture ap logs through tftp. If cp dump logs need to be captured, the -a parameter cannot be used. If capturing CP logs on PCIe devices, use – a 1;
9	-i	No	Specifies the IP address of the UDX710/ SL8563/Qualcomm/Samsung FTPlog fetching function. For example:./ logtool -i 10.20.3.55 -u admin -w 123456， Among them, 10.20.3.55 is the IP address of the FTP server.
10	-c	Yes	Required when capturing the Samsung platform CP logs, to identify the mode as CP log capture. You must add 2 after the -c parameter, for example, -c 2.
11	-e	Yes	In the PCIE environment, you must set this parameter when capturing dump logs. This parameter indicates that the pcie channel is used and the tcp protocol is used.
12	-u	No	Specifies the user name for the FTPlog capture function of the UDX710/ SL8563/Qualcomm platform. For example:./logtool -i 10.20.3.55 -u admin -w 123456， Where admin is the username of the FTP server
13	-w	No	Specifies the password of the FTPlog capture function of the UDX710/ SL8563/Qualcomm platform

			For example:./logtool -i 10.20.3.55 -u admin -w 123456， Among them, 123456 is the password for the FTP server
14	-P	No	Specifies the port for grabbing logs in proxy mode For example:./logtool –P 2500， Among them, 2500 is the port number of the proxy
15	-t	No	Specify the platform for logtool The platform number is as follows: 1: ZTE V3E platform 2: ZTE V3T platform 3: Qualcomm Platform 4: Unisoc Udx710 Platform 5: Unisoc 8850/8910 Platform 6: Eigencomm platform 7: Unisoc SL8563 platform For example:./logtool -t 6， Among them, 6 represents the eigencomm platform, and this command represents getting module logs in the eigencomm platform
16	-g	No	Whether to include tcpdump logs of the cp side when the Samsung platform captures cp logs. The value can be 0 or 1. 0 indicates that no tcpdump logs are included, and 1 indicates that tcpdump logs are included.

1. When the UDX710 (FG650/FG652/FM650)/Qualcomm platform logs through FTP, the remote PC needs to run the FTPserver tool, such as the Xlight tool (download address: http://www.xlightftpd.com/download.htm) toolkit added Xlight tools installation package at the same time, also can be installed directly.
2. The Qualcomm/Samsung platform uses the tftp tool to capture dump logs. Run the tftp tool on the remote PC, the tftp tool is also added to the tool package. You can install and run it directly.
3. A port must be specified for log capturing of **PCIE** devices on the UDX710 or Qualcomm platforms, for example:

- To capture Qualcomm platform logs, run **-d /dev/mhi_DIAG**
- Command to capture dump logs on a Qualcomm platform: **-d /dev/mhi_SAHARA**
- To capture a UDX710 platform cp ap log: **-d /dev/sdiag_nr -p /dev/slog_nr**
- Grab the Samsung ap log: **-d /dev/ttyMD_ap0**
- Grab the Samsung cp log: **-d /dev/ttyMD_dm0**

4. Proxy fetching command run:

The value range of the -P parameter is less than 9000 or greater than 9999. At the same time, you must use the -d parameter to specify the port to be proxy.

Example: **./logtool -d /dev/ttyusb0 -p 8888**

5. At present, only the logtool of ZTE Platform, Qualcomm Platform, and Eigencomm Platform support compilation in Android environment. Other platforms do not support, and the adaptation of Android platform currently only supports Android 11 and below versions.
6. The applicable scenario for -t parameter is when multiple modules from different platforms are connected to the same upper computer.
7. When uploading logs via TFTP, if the logs are not stable and high-frequency output, the connection timeout will occur and the logs cannot be uploaded, please set the timeout on the tftpd64.exe side to 100.

   tftpd64.exe->Settings->TFTP->TFTP configuration ->Timeout(seconds): 100

   tftpd64.exe->Settings->TFTP->TFTP configuration ->Max Retransmit: 100

5 Some platform log capture introduction

5.1 Qualcomm platform

There are two scenarios for capturing logs on the Qualcomm platform:

1. QXDM logs in the normal state of the module (using the DIAG port of the module)
2. The dump log of the module is in the dump state abnormally. (Port type of the module is 05c6900e)

5.1.1 QXDM log capture mode in normal module state

In the normal state of the module, there are three ways to capture logs:

1. Local grab
2. FTP capture
3. Proxy fetching

The proxy and FTP modes are applicable to scenarios where the host computer space is insufficient and logs cannot be stored for a long time.

5.1.1.1 Local grab

The local grab command is as follows:

```
./logtool
```

> If "./logtool: permission denied" is displayed, run the chmod 777 logtool command
>
> To modify file permissions.
>
> If the message "cannot execute binary file:Exec format error" is displayed, the application tool is not suitable for the system. Generally, it is necessary to recompile the tool source code on this system to generate the "logtool" application tool.

5.1.1.2 FTP grab

FTP fetching methods are as follows:

1. After installing xlight, open the xlight program and enter the main interface of the program, as shown in the figure.

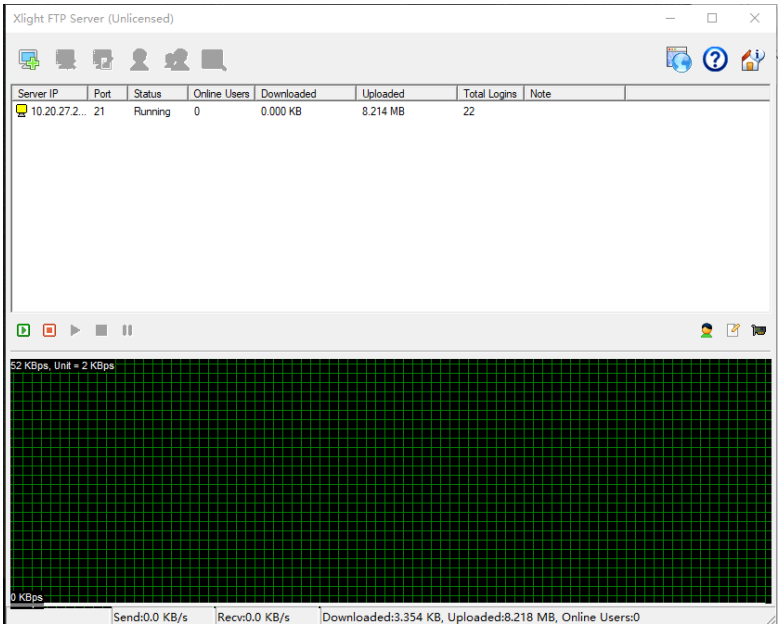

Figure 6. Xlight main screen

2. In the blank area of the main program, click the right mouse button, and select New Virtual Server from the menu that appears, as shown in the figure.

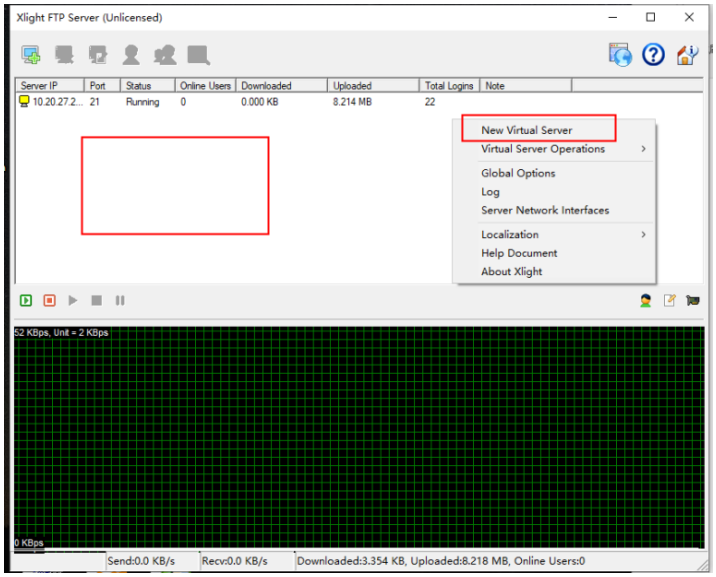

Figure 7. Select New Virtual Server

3. Run the ipconfig command to view the IP address.
4. Select the IP address obtained by ipconfig, as shown in the figure

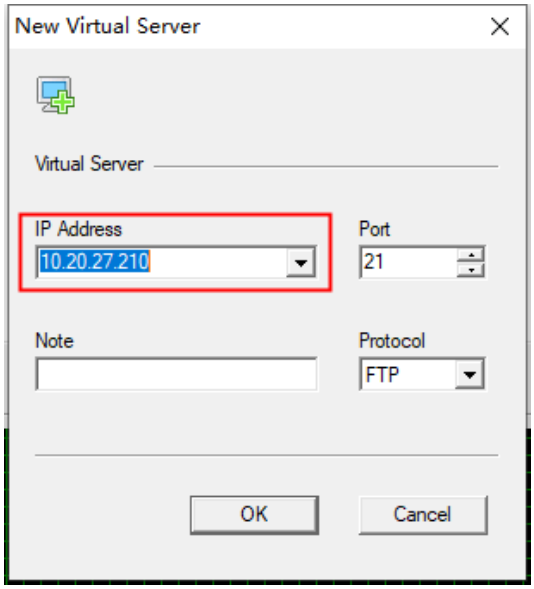

Figure 8. Select IP

5. To create an FTP account, select the created Virtual Server on the main interface of the program, as shown in the figure:

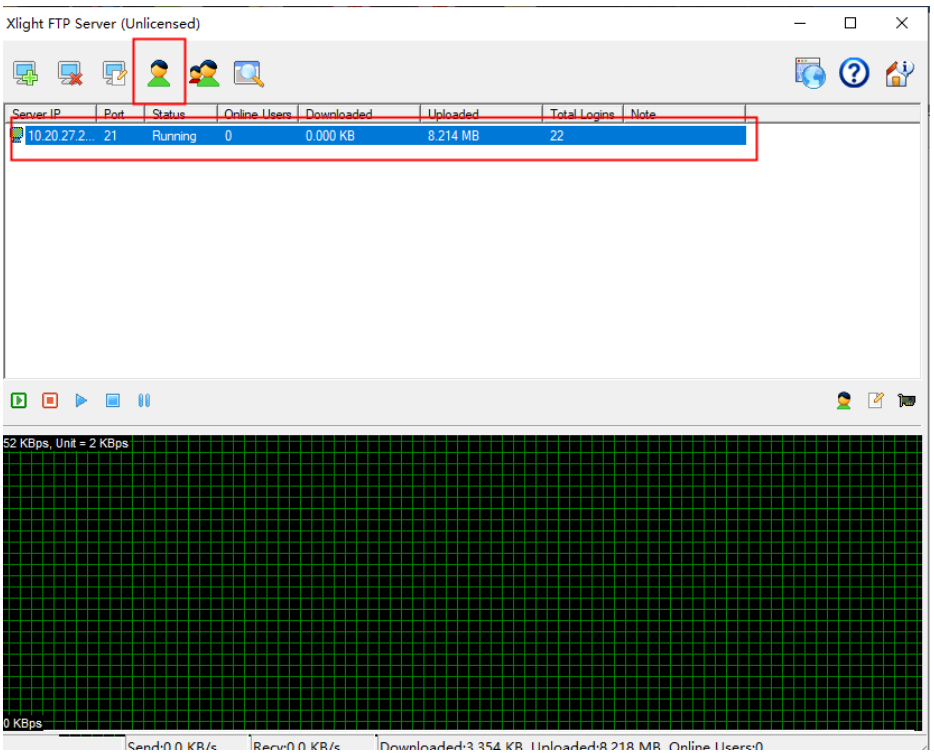

Figure 9. Select the Virtual Server to be created

6. Enter the user name, password, and ftp home directory, as shown in the figure

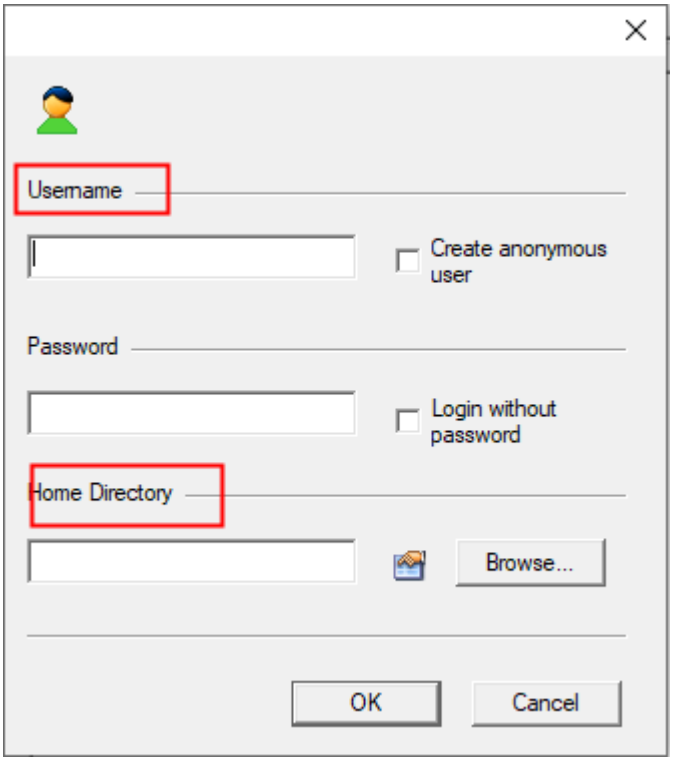

Figure 10. Username password

7. Select the User Path and double-click the selected part as shown in the figure.

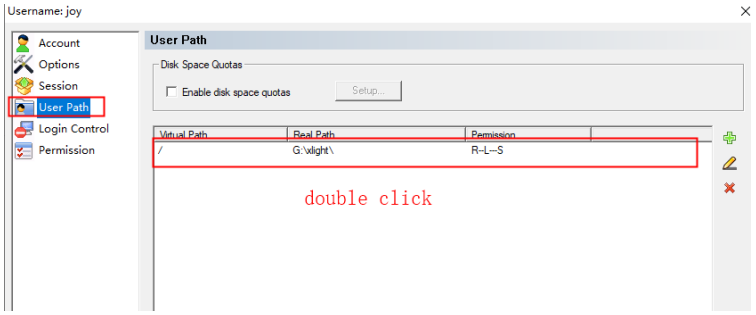

Figure 11. Select User Path

8. Select Write and Create, as shown in the figure:

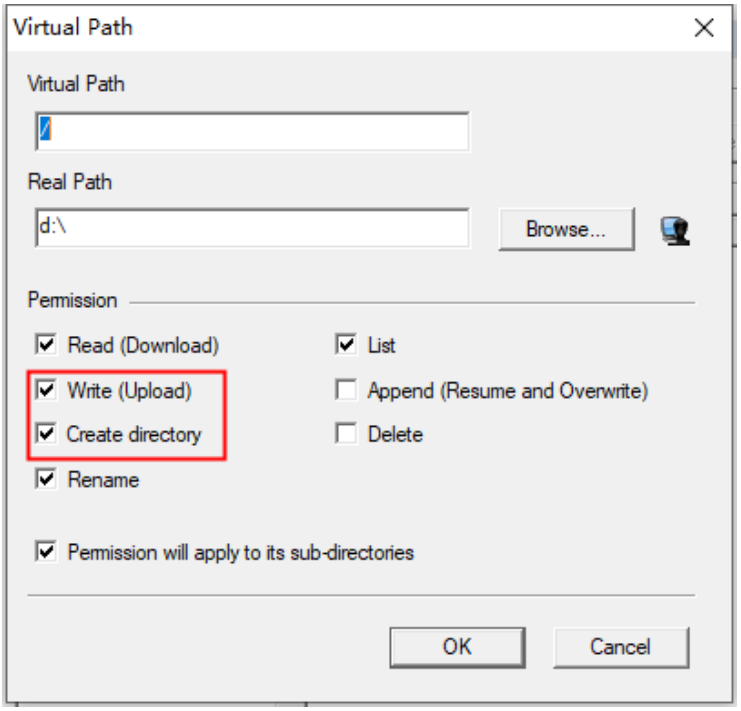

Figure 12. Select Write and Create

9. The Xlight ftp server is set up.

10. The UDX710 platform uses the ftp server to capture and execute the following command:

```
./logtool  -i 10.20.3.55 -u admin -w admin -m 50 (For details, see Chapter 4).
```

5.1.1.3 Proxy fetching

Fetching in proxy mode Prerequisites:

1. A windows PC with the QUTS/QXDM tool installed
2. Ensure that the upper computer connected to the module can ping through windows.

Specific operations are as follows:

1. Run the proxy command:

```
./logtool –d /dev/ttyUSB0 –P 2500
```

2. Configure the agent port on the windows PC

   a. Open the QUTS tool and perform the following operations. The IP address is the IP address of the upper computer connected to the module, and the port is the port specified by the execution agent tool –P.

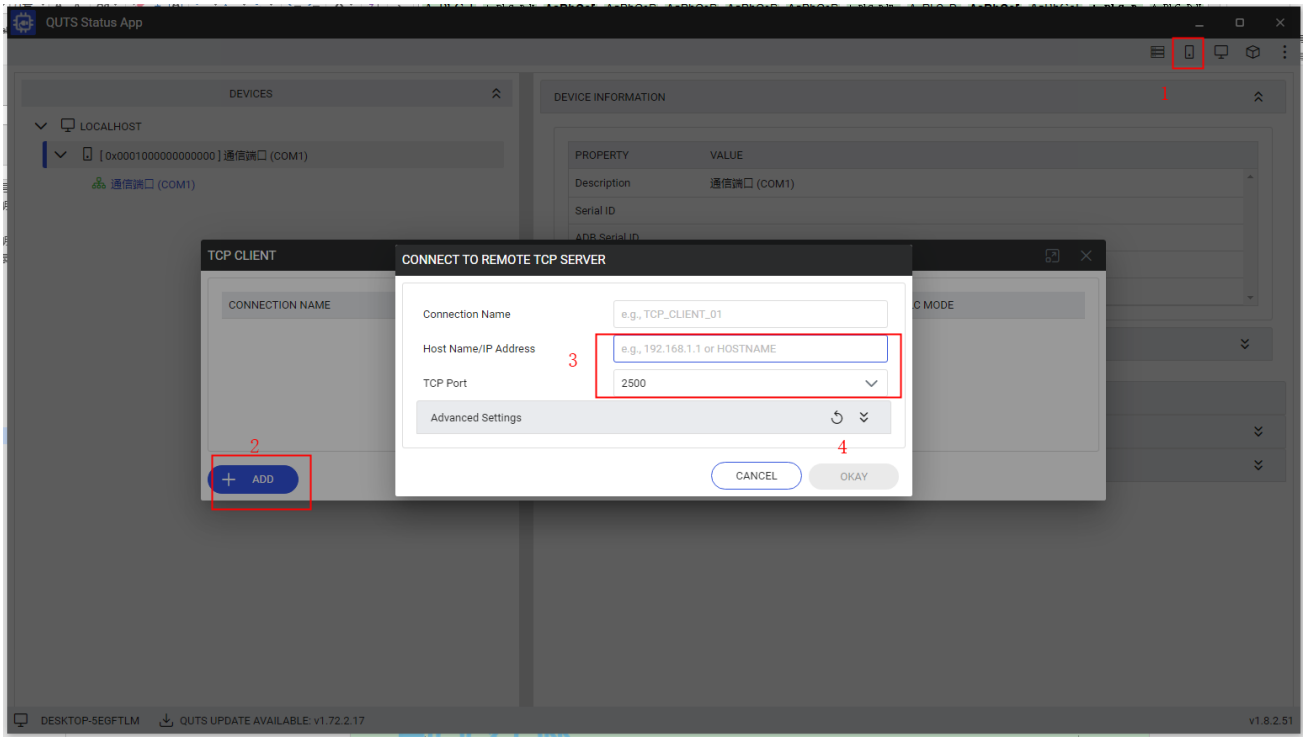

Figure 13. QUTS configuration process

b. After adding the proxy port, open the QXDM tool and you can see the added port. Click connect as shown in the figure below to capture the log.

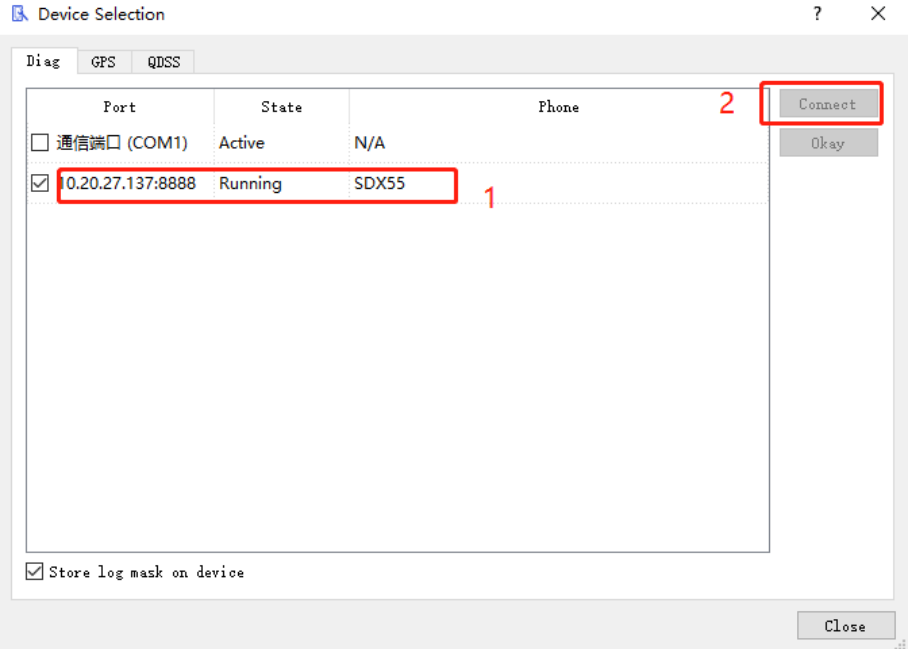

Figure 14. QXDM port connection

5.1.2 Dump log grab

There are two ways to grab Dump logs:

1. When the storage space of the client's host computer is sufficient, it can be stored directly to the host computer;

The command is as follows:

```
./logtool
```

> To use the local crawl dump function, you need to change config_qcom_localdump = no in the Makefile to config_qcom_localdump = yes. This switch is turned off by default because this function depends on the zlib open source library and is not supported by some customer environments. This results in a compilation error.

2. If space on the host is insufficient, use the tftp tool to save the dump log to a PC with large space.

The specific methods are as follows:

After installing the tftp tool, open the main interface, as shown in the following figure:

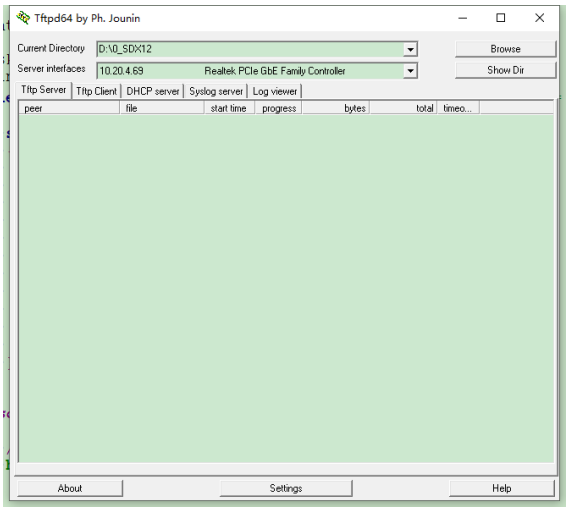

Figure 15. Tftp main interface

In the command, Current Directory indicates the log saving path and Server interfaces indicate the local IP addresses. After the configuration of the current Directory and server interfaces, you can run dump log capture commands on the Qualcomm platform on the upper computer connected to the module:

```
./logtool -i tftp:10.20.4.69
```

> Ensure that the IP address of the PC that connects to the module and the PC that runs the tftp tool can ping through each other

5.2 UDX710/SL8563 platform

There are four scenarios for log capture on UDX710/SL8563 platform:

1. AP logs and CP logs of the module in normal state
2. The module enters the dump state of ap dump and cp dump abnormally

5.2.1 AP log grab

There are two ways to capture AP logs:

1. Local capture of logs

When the storage space of the client's host computer is sufficient, it can be stored directly to the host computer;

The command is as follows:

```
./logtool -a 0
```

2. Tftp captures logs

If the space of the upper computer is insufficient and the module needs to be suspended for a long time, use the Tftp tool to save the ap log to a PC with large space. Double-click tftpd64.exe to configure the log storage path and IP address (IP address in Windows).

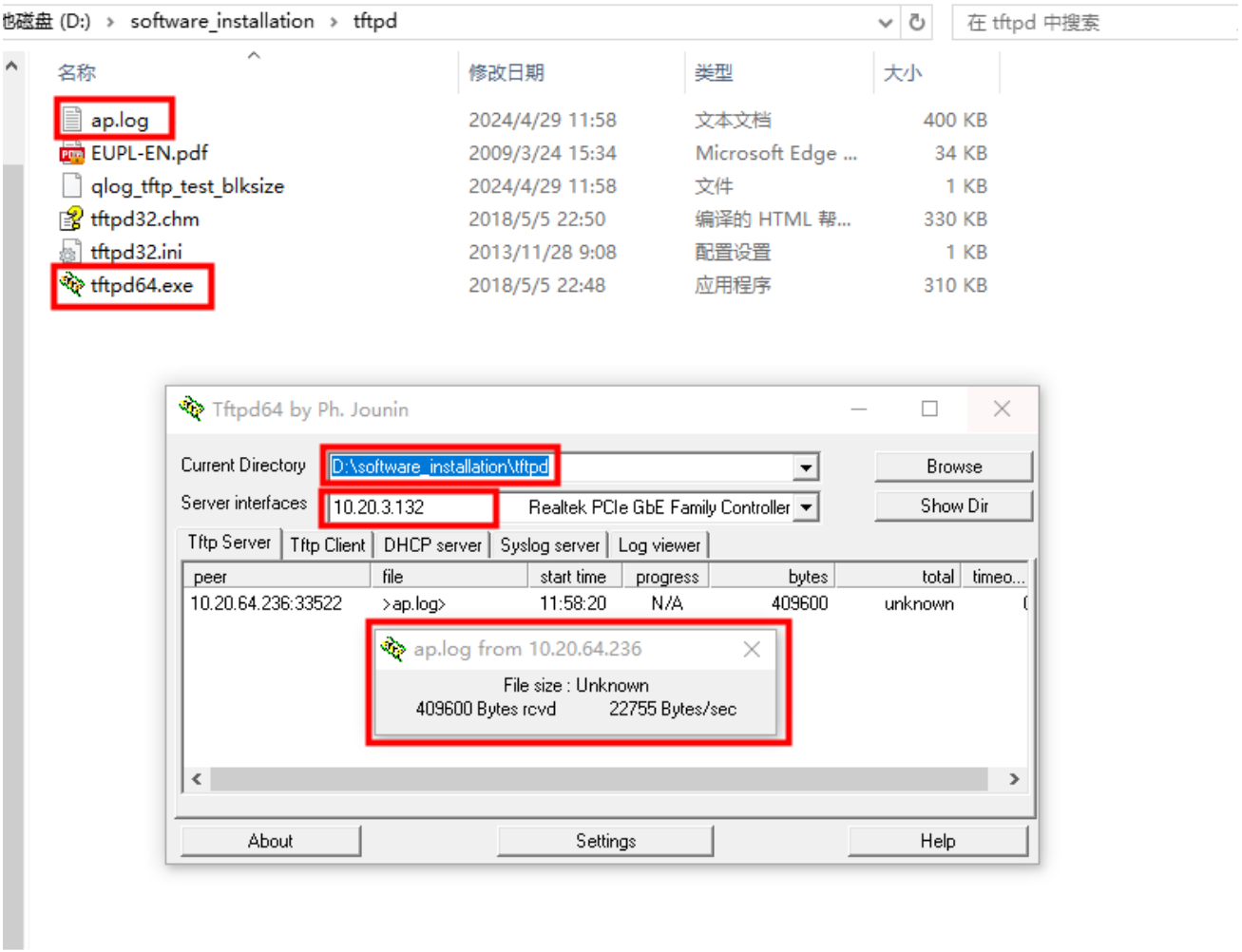

Figure 16. Tftp grab ap log

After the configuration is complete, run the following commands on the host computer to capture ap logs, as shown in the ap.log file in the above figure:

```
./logtool –i tftp:10.20.3.132 –a 2
```

Ensure that the IP address of the PC that connects to the module and the PC that runs the tftp tool can ping through each other

5.2.2 CP log grab

CP log fetching can be done in two ways:

1. Local capture of logs

When the storage space of the client's host computer is sufficient, it can be stored directly to the host computer;

The command is as follows:

```
./logtool -m 30 –n 3 –s logfile
```

-m/-n/-s parameters can be set as required. For details, see Section 4.

2. Proxy method for capturing logs

If space on the host is insufficient, run the logtool+Logel command to save CP logto the PC in the Windows operating system. The PC and the host are connected through a network cable.

The command is as follows:

```
./logtool –d diag_port –P 8888 &

./logtool –d log_port –P 8889 &
```

Send the following AT command before grabbing, otherwise there may be a significant loss of logs.

```
AT+SPLOGLEVEL=1,3,"60FBF8700FFD0008FFF0000000000000","E7F302E708FFFFFFFFFF70E7CCBFF1
C000FF030CFDFFFFFF8F710000000000E3"
```

In Windows, you need to install the Logel tool provided by Zhanrui platform. Open the tool, click "capture setting" in the upper left corner, and the pop-up box is shown as follows.

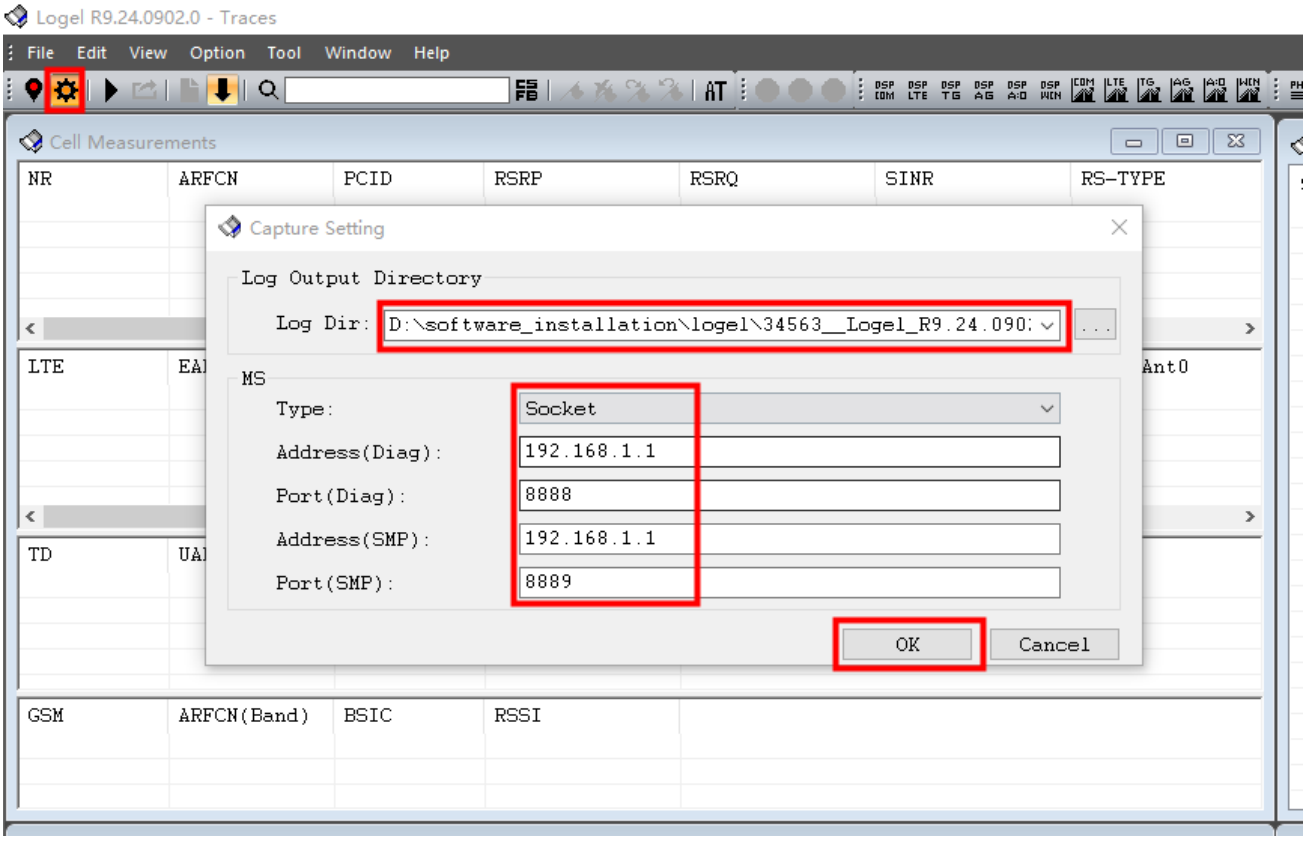

Figure 17. logtool+Logel grab cp log

Log Dir: log storage location. You can set the location based on site requirements.

Type: Select the Socket mode.

Address(Diag) : indicates the IP address of the host

Port(Diag) : indicates the proxy port corresponding to the diag port

Address(SMP) : indicates the IP address of the host

Port(SMP) : indicates the proxy port corresponding to the log port

3. FTP method capture, refer to section 5.1.1.2 for usage instructions.

In scenarios where there is insufficient space on the upper computer, it is generally recommended to use the proxy method, but not the FTP method. This is because the unicom platform has a large amount of logs, which can easily result in the next log file being captured when the current file is not sent through FTP, causing the memory to be full and the upper computer to be unable to use it normally.

5.2.3 AP dump log grab

Before capturing AP dump, the Sciu2s driver must be integrated on the host computer. For the integration method, refer to Chapter 5 of the document "Fibocom_F6 Series_USB Driver Installation Guide_Linux".

There are two ways to capture AP dump:

1. Local crawling

When the client's upper computer has sufficient storage space, it can be directly fetched to the local storage

```
./logtool
```

2. TFTP mode capture

Note You need to use the tftp tool to save AP dump logs to a PC with large space. Double-click tftpd64.exe to open the tftp tool and configure the log storage path and IP address (IP address in Windows).

If you run the following command on the host, the dump logs are stored in the configured path:

```
./logtool -i tftp:10.20.3.132
```

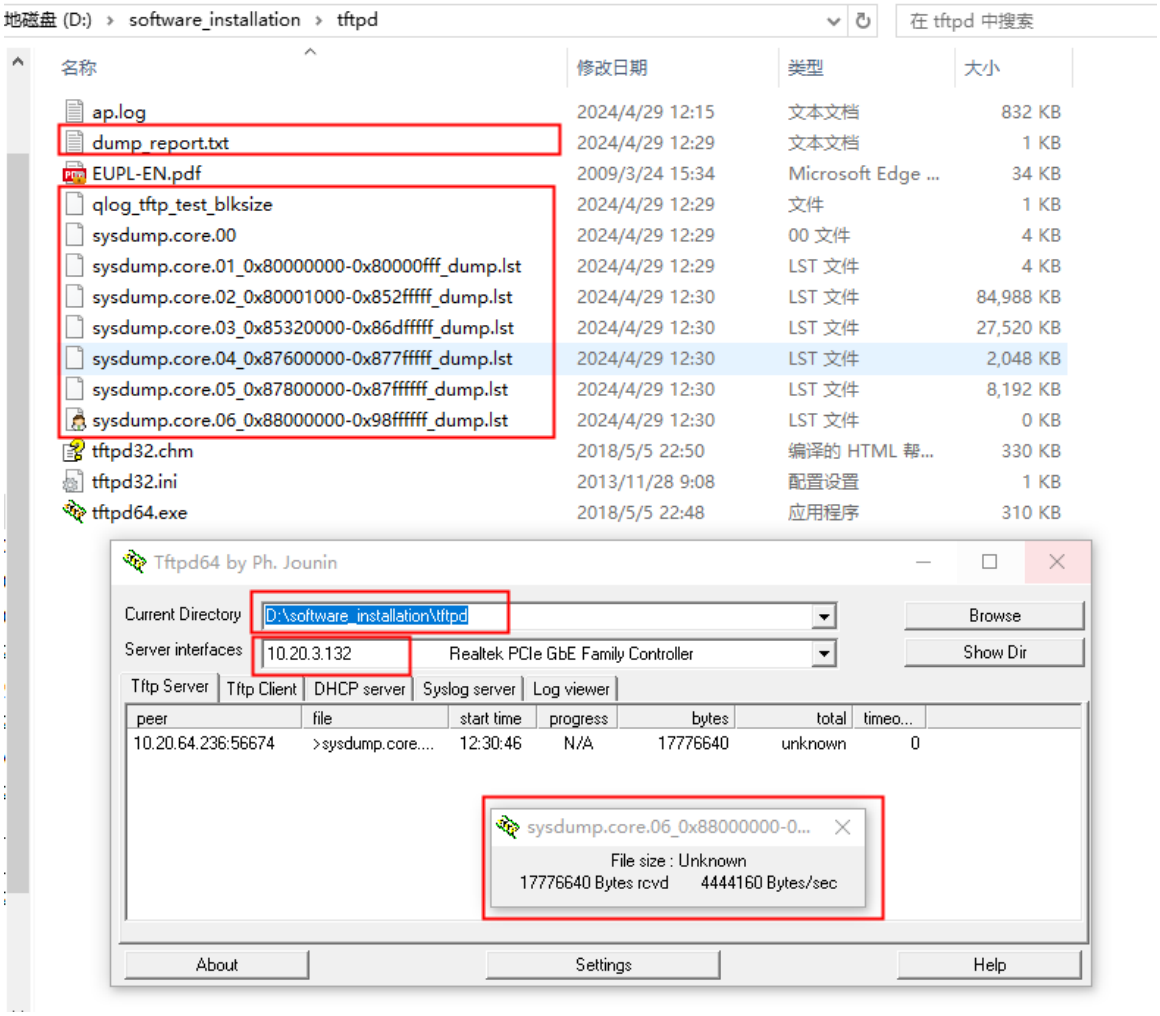

Figure 18. tftp grab ap dump

5.2.4 CP dump log grab

CP dump log capture supports two methods

1. Local capture, suitable for scenarios with sufficient space on the upper computer;

2. Proxy method (logtool+Logel) capture, suitable for scenarios where the upper computer space is insufficient;

> CP dump logs are large and the host space is generally insufficient. Therefore, logtool+Logel is recommended to capture CP dump logs.
>
> For details, refer to the second method in Section 5.2.2.

5.3 Spreadtrum Platform (PCIe)

The ports enumerated by PCIe devices are different. Therefore, you need to specify ports for log capturing and meet the following prerequisites:

Example:

- host computer's **diag** port name and path: **/dev/sdiag_nr**
- host computer's **log** port name and path: **/dev/slog_nr**
- The IP address of the current device is 192.168.0.1.
- The TFTP address is 192.168.0.2.

Prerequisites:

1. The **log** port of the PCIe device spits out logs. Check method:

   Send the **at+gtlogen=1** command to the module. Run **cat /dev/slog_nr** and you can find logs in this path.

2. logtool has been cross compiled on customer's compilation chain.

5.3.1 AP Log Capture

You can choose an appropriate log capturing method based on the log capturing time and the memory space of customer's host computer.

- TFTP capture

```
./logtool -d /dev/sdiag_nr -i tftp:192.168.0.2 -a 2
```

For other TFTP settings, see section 5.2.1.

5.3.2 CP Log Capture

You can choose an appropriate log capturing method based on the log capturing time and the memory space of customer's host computer.

1. Local capture

```
./logtool -p /dev/slog_nr -a 1
```

2. TFTP capture

```
./logtool -d /dev/sdiag_nr -p /dev/slog_nr -i tftp:192.168.0.2
```

3. Proxy capture

```
./logtool -d /dev/sdiag_nr -P 8888 &
./logtool -p /dev/slog_nr -P 8889 &
```

If a dump occurs during TFTP and proxy capturing, you can capture CP dump log at the same time.

For other settings of the preceding methods, see section 5.2.2.

5.3.3 Capture AP and CP Logs At the Same Time

Due to the limitations of the Unisoc log tool, CP dump log capture requires two ports. Therefore, when a dump occurs, AP logs and CP logs cannot be captured at the same time. The CP dump file needs to be transmitted over the **diag** port. The port is occupied and data cannot be transmitted.

If a dump does not occur, capturing CP logs requires sending an AT command, which occupies the **diag** port, and TFTP cannot transmit data of two channels at the same time. Therefore, dump must not happen during simultaneous capturing, and only the following two methods are supported:

1. Local capture

```
./logtool -d /dev/sdiag_nr -p /dev/slog_nr -a 0
```

2. Local capture for AP logs and TFTP capture for CP logs

```
./logtool -d /dev/sdiag_nr -p /dev/slog_nr -i tftp:192.168.0.2 -a 0
```

5.4 Samsung platform

For details, see *Fibocom_F5 Series_Log Capture Guide_Linux*. The following is only a simple example.

1. Usb-based mode

- Captures AP logs

```
./logtool -d /dev/ttyUSB1
```

- Captures CP logs

```
./logtool -c 2 -d /dev/ttyUSB2 -g 1
```

- Captures DUMP logs

```
./logtool -i tftp:IP
```

2. In PCIE mode

- Captures AP logs

```
./logtool -d /dev/ttyMD_ap0
```

- Captures CP logs

```
./logtool -c 2 -d /dev/ttyMD_dm0 -g 1
```

- Captures DUMP logs

```
./logtool -e tcp -i tftp:IP
```